

Specifying AeST's free function: one no-go, one live candidate

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Aether–scalar–tensor gravity (AeST) [Skordis & Złośnik, Phys. Rev. Lett. **127**, 161302 (2021)] is the only relativistic MOND-class theory known to reproduce the cosmic microwave background power spectrum, and it contains a single free function $\mathcal{F}(\mathcal{Y}, \mathcal{Q})$ that its authors leave unspecified. We show that specifying it is obstructed, and we extract from the obstruction three necessary conditions on *any* relativistic completion of MOND phenomenology. Each is established by an explicit construction that satisfies the previous conditions and fails the next. **(R1)** The free function must depend on the gradient of the *total* potential, not on the scalar's own. With $\mathcal{F}(\mathcal{Y})$ the quasi-static reduction forces the anomalous acceleration $U(y)$ to be monotone increasing, hence to saturate at some $U \rightarrow s$, hence to produce a *constant* sunward anomaly $s a_0$ at every planetary distance. Perihelion precession then requires $s \leq 1.27 \times 10^{-5}$ while the radial acceleration relation requires $s \geq 0.435$ — incompatible by $1.2\text{--}3.4 \times 10^4$. There is no external-field relief: computed for this family rather than imported, it is $1.000000\times$. **(ii)** Replacing $\mathcal{Y}$ by $Z \equiv J^\mu J_\mu$, the squared aether acceleration, satisfies R1 exactly: the quasi-static reduction becomes AQUAL, the exponential interpolation becomes legal, the 1 AU anomaly falls to $10^{-3458.7} \text{ ms}^{-2}$, and the gap is voided. This construction is a *live candidate*, not a casualty. Its Friedmann–Robertson–Walker (FRW) background is exactly undisturbed — $\bar{Z} = 0$ identically for arbitrary lapse, so the Friedmann equations, $w = -1$, the dust mode, $Q(a)$ and the derived $a_0(z)$ law are unchanged — and its analytic linear-in- Z piece is a mere relabelling $K_B \rightarrow K_B - \mathcal{F}_Z(0)$ of the aether's kinetic coefficient. Its one real cost is that, eating the total gradient, its MOND branch is switched *on* at recombination, where the acoustic-band gravitational acceleration is only $7.6\text{--}11.9 a_0$ absent running; the resulting $\sim 5\%$ distortion of linear gravitational dynamics is worth $18\text{--}112\sigma$ at unit response. It survives only through $a_0(z)$ suppression — and the suppression the framework independently derives, $a_0(\text{rec})/a_0(0) = 0.0060$, meets the requirement ($\leq 9.7 \times 10^{-3}$ to 3.7×10^{-1}) on every one of eight forks. **(iii)** The residual risk is priced rather than hidden: delivering that suppression needs $\nu_0 \geq 5.6 \times 10^{-9}$ to 8.2×10^{-6} , and two of eight forks put this above the radial-acceleration-relation ceiling 2.36×10^{-6} — exactly the $\alpha = 1$ kernel under the conservative reading, by $2.4\text{--}3.5\times$, while the exponential kernel that this construction re-legalises clears on every reading. The whole interlock is hostage to one uncomputed number, the response $S = |d \ln C_\ell / d \ell|$: every fork clears for $S \leq 0.54$, every fork clashes for $S \geq 20.6$. **(iv)** We also show what *cannot* rescue R1: a rescaling $\mathcal{J} \rightarrow s\mathcal{J}$ forces $\bar{G}/G_N \leq 0.121$, giving a primordial helium abundance $Y_p = 0.086$ against an observed 0.2449 ± 0.0040 (-42σ) and a gravitational radiation density *below that of the photons alone*. Gravitational waves do not pay this price: the standard-siren amplitude's sensitivity is exactly zero. We also record a structural identity: AeST's action already contains an explicit $+(2 - K_B)a^\mu a_\mu$, so its apparent $c_4 = 0$ is an artifact of variables; and $c_T = 1$ holds exactly, for any free function.

I. SETUP

AeST's action, verified against the authors' source [1, 2], is

$$S = \int d^4x \frac{\sqrt{-g}}{16\pi\bar{G}} \left[R - 2\Lambda - \frac{K_B}{2} F^{\mu\nu} F_{\mu\nu} + 2(2 - K_B) J^\mu \nabla_\mu \varphi - (2 - K_B) \mathcal{Y} - \mathcal{F}(\mathcal{Y}, \mathcal{Q}) - \lambda(A^\mu A_\mu + 1) \right] + S_m[g], \quad (1)$$

with $\mathcal{Q} = A^\mu \nabla_\mu \varphi$, $\mathcal{Y} = (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi$, $F_{\mu\nu} = 2\nabla_{[\mu} A_{\nu]}$, and $J^\mu = A^\nu \nabla_\nu A^\mu$ the aether's acceleration. Matter couples to $g_{\mu\nu}$ alone.

Throughout we use the acceleration scale

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad (2)$$

with an alternative footing giving 1.1279×10^{-10} ; both are carried throughout. The coefficient $\kappa = \frac{1}{2}$ is *fit-*

ted, not derived: three determinations give 0.551 ± 0.043 , 0.465 ± 0.076 and 0.537 ± 0.071 , combining to 0.529 ± 0.034 . Write $y = g_{\text{bar}}/a_0$ and $U(y) = u/a_0$ for the anomalous acceleration.

II. R1: THE FREE FUNCTION MUST EAT THE TOTAL GRADIENT

With $\mathcal{F}(\mathcal{Y})$, the quasi-static gradient sector diagonalises: $\Psi = \Psi_N + \chi$ with $\nabla^2 \Psi_N = 4\pi\hat{G}\rho$, and in spherical symmetry Gauss's theorem gives the local algebraic law $u J_Y(u^2) = g_{\text{bar}}$ with $u = |\nabla\chi|$. The scalar's own gradient *is* the anomalous acceleration. Reconstructing the free function from a chosen interpolation gives $J_Y(Y) = y(U)/U$, invertible iff U is strictly increasing; equivalently the longitudinal eigenvalue of $M_{ij} = J_Y \delta_{ij} + 2J_{YY} u_i u_j$ is positive under the same condition. A non-monotone U means both a multi-valued $\mathcal{F}$

and a longitudinal gradient ghost.

This is specific to AeST. In AQUAL [3] the free function depends on the total potential's gradient and stability requires only $dg_{\text{obs}}/dg_{\text{bar}} > 0$ — satisfied by the exponential interpolation $\nu = 1/(1 - e^{-\sqrt{y}})$ [4], minimum 0.968. The same kernel is legal in one theory and fatal in the other, purely from which gradient the free function eats.

Monotonicity plus $U/y \rightarrow 0$ forces saturation, $U \rightarrow s$: a constant sunward anomaly $s a_0$ at every planetary distance. By the Gauss planetary equations a constant radial perturbation gives $\dot{\omega} = s a_0 \sqrt{1 - e^2}/(na)$; against anomalous-precession limits [5, 6] the worst planet requires

$$s \leq 1.27 \times 10^{-5} \text{ (canonical), } 1.05 \times 10^{-5} \text{ (alt).} \quad (3)$$

Applying $U(2) \geq 0.4$ to the family $J_Y = v/(1 - v/s)$, $v = \sqrt{Y}/a_0$, gives $s \geq 0.4348$; relaxing to a global fit (rms ≤ 0.15 dex on 3389 SPARC points [7], Υ in the Spitzer band) gives 0.157–0.294 depending on treatment. Hence

$$\boxed{\text{GAP: } 1.2\text{--}3.4 \times 10^4.} \quad (4)$$

Every candidate relief was computed and none helps. *The external-field effect contributes nothing:* derived for this family by two independent routes (an $\ell = 1$ penetration ODE and a flux bound requiring no perturbation theory), the relief on the saturated anomaly is $1.000000 \times$ — the external field screens *itself*, not the anomaly, and somewhere on the 1 AU sphere the anomaly is at least $s a_0(1 - 4 \times 10^{-9})$. A density-dependent a_0 *hurts*: with suppression f the constrained product is sf , and $f U_s(2/f) = 0.4$ gives 0.435 at $f = 1$ but 2.00 at $f = 0.1$, unsatisfiable below $f = 0.080$ because $U \leq \sqrt{y}$ caps the family. Solving the non-spherical problem for a Miyamoto–Nagai disc raises the required floor by 1.07–1.10. Refitting Υ buys 1.34 $\times$.

III. THE Z-FORM: A LIVE CANDIDATE, AND ITS PRICED COST

R1 says: use the total gradient. AeST already contains the object that supplies it. With $D^\mu = q^{\mu\nu} \nabla_\nu \varphi$ one has the identity, verified three times independently and requiring no computer algebra,

$$\begin{aligned} & 2(2 - K_B)J \cdot \nabla \varphi - (2 - K_B)\mathcal{Y} \\ & \equiv +(2 - K_B)Z - (2 - K_B)|D - J|^2, \end{aligned} \quad (5)$$

with $Z \equiv J^\mu J_\mu$. AeST therefore already contains an explicit $+(2 - K_B)a^\mu a_\mu$: its apparent $c_4 = 0$ is an artifact of variables. Moreover $J^i = \nabla^i \Psi$ with coefficient exactly unity, with Z 's leading coefficient independent of φ , Φ and the aether perturbation — so Z is the total potential gradient squared.

Replacing $\mathcal{F}(\mathcal{Y}, \mathcal{Q})$ by $\mathcal{F}(Z, \mathcal{Q})$ then works exactly as R1 requires. The aether-longitudinal equation becomes $v = \nabla \Psi$ pointwise, the $(1 + J_Y)$ factor disappears, and the reduction is *AQUAL exactly*, with $\mu(g_{\text{obs}}) = J_Z(g_{\text{obs}}^2)$ and $\hat{G}$ unchanged. The exponential kernel becomes legal (min $dx/dy = 0.968$, against min $dU/dy = -0.032$ under $\mathcal{F}(\mathcal{Y})$ for the same kernel), and the 1 AU anomaly becomes $10^{-3458.7} \text{ m s}^{-2}$ — some 10^{3445} below the Sereno–Jetzer bound [8], replacing an $s a_0$ that sat $1279 \times$ above it. Further, $c_T^2 = 1$ exactly for any free function, and G_N is finite and equal to the unmodified value.

This construction is not a casualty; it is a *live candidate*, and its cosmology has been computed rather than assumed [14].

The background is exactly undisturbed. On FRW the comoving aether congruence is geodesic, so $J^\mu = 0$ identically for arbitrary lapse $N(t)$, hence $\bar{Z} = 0$ and $\mathcal{F}(0, \mathcal{Q}) = K(\mathcal{Q})$ — the same background function AeST's $\mathcal{F}(\mathcal{Y}, \mathcal{Q})$ reduces to. The new term's stress-tensor contribution carries at least one factor of J and therefore vanishes too. Friedmann, $w = -1$, the dust mode, $Q(a)$, and consequently the derived $a_0(z)$ law are bit-for-bit the AeST objects.

At linear order the analytic piece is a relabelling, not a degeneration. One finds $J_i = \nabla_i E$ exactly, with $E = \dot{\alpha} + \Psi$ the theory's own mixing variable, so $Z = |\nabla E|^2/a^2$ is *second* order. Since $F^{\mu\nu} F_{\mu\nu} = -2Z$ at $O(\delta^2)$, any analytic linear-in- Z piece of $\mathcal{F}$ is nothing but a shift $K_B \rightarrow K_B - \mathcal{F}_Z(0, \mathcal{Q})$ — and on the determined family $\mathcal{F}_Z(0) = (2 - K_B)\mu_{\text{phys}}(0) = 0$, because the deep-MOND end of every interpolation has $\mu_{\text{phys}} \rightarrow 0$. (The degeneracy is scalar-sector only: a transverse aether mode sees a magnetic $|\nabla \times A^V|^2$ piece that Z does not contain, so a nonzero $\mathcal{F}_Z(0)$ would change the vector dispersion relation and must be re-checked separately.)

The real cost. Because the free function now eats the total gradient, its MOND branch switches on wherever the total acceleration nears a_0 — and a CLASS run puts the acoustic-band gravitational acceleration at recombination at only $y = 7.6\text{--}11.9$ in units of a_0 , absent running. The fractional modification $\epsilon = 1 - \mu(y) \simeq 1/2y$ is then a few per cent, kernel-robustly, and *not* exponentially small. The $\mathcal{Y}$ -form is immune only because its MOND term is $O(\delta^3)$ at any acceleration. Aggregated over $\ell = 2\text{--}2500$ at $f_{\text{sky}} = 0.7$ and unit response, this is $18\text{--}112\sigma$ across eight forks (footing $\times$ kernel $\times$ per-mode/total-field). Truncating at $\ell \leq 1600$, where Planck TT is not noise-dominated, still leaves 47σ on the reference fork — the conclusion does not rest on the noise-dominated tail.

And the framework's own law pays it. The CMB tolerates the Z -form only if $a_0(\text{rec})/a_0(0) \leq 9.7 \times 10^{-3}$ to 3.7×10^{-1} across the eight forks. The suppression this framework independently derives is 0.0060, which clears the tightest fork and the loosest. This is the single most favourable structural fact about the construction: the suppression it needs and the suppression the theory already predicts are the same order, and neither was tuned

to the other.

The residual risk, priced. Delivering that suppression requires $\nu_0 \geq 5.6 \times 10^{-9}$ to 8.2×10^{-6} . Two of the eight forks put this above the independent RAR ceiling $\nu_0 \leq 2.36 \times 10^{-6}$, by $2.4\text{--}3.5\times$ — and they are exactly the $\alpha = 1$ kernel under the conservative per-mode reading. Six clear with room. This matters because the Z -form *reopens the kernel choice in its own favour*: the legality theorem of Sec. II, which selected $\alpha = 1$ in the $\mathcal{Y}$ -form, does not apply here, so the exponential kernel is legal again — and it clears the ceiling on every reading, differing from $\alpha = 1$ by $4\times$ in aggregate (17.6 vs 36.7 σ) because $\mu \rightarrow 1$ exponentially rather than as $1 - 1/2y$.

What is owed. The whole interlock scales as $\nu_0^{\text{req}} \propto S^2$ in the CMB’s fractional response $S = |d \ln C_\ell / d\epsilon|$, which is *not computed*: every fork clears for $S \leq 0.54$, every fork clashes for $S \geq 20.6$, and the $S = 1$ used above is an assumption, not a result. Settling it requires a Boltzmann code carrying A_μ , the constraint, and $\mathcal{F}(Z, \mathcal{Q})$ — a build, not an argument. A second approximation is also adverse and named: the free function eats ∇E , and $E \sim \Psi$ was used as the subhorizon proxy; if $E < \Psi$ the distortion is *larger*.

IV. WHAT CANNOT RESCUE R1: THE $\tilde{G}/G_N$ SPLIT

We record for completeness a modification that does *not* work, since it is the natural thing to try and its failure is instructive. Both sector conditions constrain only the total Z -coefficient: static attraction needs it positive, vector no-ghost needs it below K_B . Rescaling the free function, $J \rightarrow sJ$, satisfies both at every acceleration for $s < K_B/(2 - K_B)$; the window is $0 < s < 0.138$ at $K_B = 0.25$, and the solar-system screening is untouched, being s -independent.

The price is that with $\mathcal{F}(Z)$ the non- $\mathcal{F}$ static Lagrangian is annihilated identically by $v = \nabla \Psi$, making the free function’s normalisation the sole source of Newton’s constant: $G_N = \tilde{G}/s$. The no-ghost condition is therefore equivalent to $2\tilde{G} < K_B G_N$, i.e.

$$\tilde{G}/G_N \leq 0.121 \quad (K_B = 0.25), \quad 0.048 \quad (K_B = 0.10), \quad (6)$$

against AeST’s 0.875 — an $8.3\times$ to $20.7\times$ split. It cannot be paid.

Big-bang nucleosynthesis. $Y_p = 0.086$ at the no-ghost ceiling against an observed 0.2449 ± 0.0040 : -42σ from the same calculation’s $\tilde{G}/G_N = 1$ control. A closure-free bound assuming no nucleosynthesis model at all still gives $Y_p \leq 0.176$, i.e. -17σ .

The CMB, and this is an inconsistency rather than a tension. The gravitational sector’s radiation density comes out at 0.2046 of the thermal photon density — *below the photons alone*, whose value FIRAS [9] fixes with no gravity anywhere in the chain. The equivalent $N_{\text{eff}} = -3.503$. The hard floor is missed by $4.9\times$ ($K_B = 0.25$)

to $12.2\times$ ($K_B = 0.10$), and the ceiling sits 139σ from viability in cosmic-variance units.

Gravitational waves do not pay this price, and we state that against our own interest. The tensor sector is exactly Einstein–Hilbert: on FRW plus a transverse-traceless (TT) mode, $J^\mu = 0$, $Z = 0$, $F_{\mu\nu} = 0$ and $\mathcal{Y} = 0$ all hold identically and to all orders in h , so the dark sector carries no anisotropic stress for any free function, giving $c_T = 1$ and $\alpha_M = 0$ exactly. The standard-siren amplitude’s sensitivity to s is then exactly zero: $\tilde{G}$ enters the strain and G_N the orbit, and they cancel in d_L^{GW} . GW170817 [10] places *no bound* on s . Two liabilities the same derivation creates are recorded but *not scored*: the inferred chirp mass rescales as $r^{3/5}$ ($4.13 M_\odot$ for GW170817 at the ghost edge, which needs the theory’s own strong-field stellar structure to close), and the tensor-channel orbital decay is $\dot{P}_b/\dot{P}_b^{\text{GR}} = r$, an $8\times$ deficit against B1913+16 [11]. We decline to score the second because the identical ledger applied to *unmodified* AeST would exclude it at 77σ — the tensor-only flux ledger is evidently incomplete, and the total radiated flux is not computed here. The usable result is a constraint on the shape of any repair: the missing channels must supply the same $1 - r$ fraction at every orbital period, since $\dot{P}_b/\dot{P}_b^{\text{GR}}$ is measured to 0.16% in a 7.75 h orbit and to 1.3×10^{-4} in a 2.45 h one [12], and a dipole enters at relative order $(c/v)^2$ and cannot be period-independent.

V. TESTABLE PREDICTIONS

A no-go is tested by construction rather than by observation, but the derivations above carry four consequences that data can decide.

P1: $\alpha_M = 0$ and $c_T = 1$, exactly, for any free function. This is the strongest prediction here and the one least dependent on anything we have chosen. On FRW plus a TT mode, $J^\mu = 0$, $Z = 0$, $F_{\mu\nu} = 0$ and $\mathcal{Y} = 0$ hold identically and *to all orders in h* , so every non-Einstein invariant is h -independent and the dark sector’s TT stress is $T^x_x - T^y_y = 0$. The tensor sector is therefore exactly Einstein–Hilbert with $M_*^2 = 1/8\pi\tilde{G}$ constant. AeST and every construction in this paper are consequently indistinguishable from general relativity in gravitational-wave propagation — no running Planck mass, no modified friction, no c_T deviation, at any redshift and for any K_B , s or a_0 footing. GW170817’s $|c_T - 1| \lesssim 10^{-15}$ is passed trivially rather than fitted. A *confirmed $\alpha_M \neq 0$ from LISA or Einstein Telescope standard sirens falsifies this entire class in one measurement*, and cannot be absorbed by any choice of free function.

P2: the orbital-decay deficit must be period-independent. If the tensor-only flux ledger of Sec. IV is incomplete, as we argue, the missing channels must supply the same $1 - r$ fraction of the flux at every orbital period. A dipole channel enters at relative order $(c/v)^2$ and therefore cannot be period-independent; the repair must renormalise the quadrupole coefficient itself. Any

future binary showing a period-*dependent* deviation from the GR quadrupole formula excludes the repair, and with it the aether-vector rescaling.

P3: the $\mathcal{F}(\mathcal{Y})$ class predicts a constant sunward anomaly with a fixed planet ordering, $\hat{\omega} = s a_0 \sqrt{1 - e^2} / (na)$, falling with semi-major axis in a way no fitted $\hat{\omega}$ per planet can mimic, since a single s sets all of them. This is already excluded (Sec. II), which is *why* R1 exists; a future ephemeris solution reporting a coherent residual of this shape would revive rather than kill it.

P4: the Z -form predicts no solar-system deviation at all, at any measurable level: the 1 AU anomaly is $10^{-3458.7} \text{ m s}^{-2}$ and is s -independent. Any claimed solar-system MOND detection excludes it outright.

These are separate from framework-level predictions that this paper leaves untouched, since they belong to the normalisation (2) rather than to any completion: a hash-frozen wide-binary velocity ratio decided by Gaia DR4; a declining $a_0(z)$, which puts the MOND regime *off* at recombination at 0.0060 of its present value and is falsified by any robust a_0 evolution below $z \sim 5$; and $\nu_0 \leq 2.36 \times 10^{-6}$, i.e. no environmental variation of a_0 at a level current data could see.

VI. WHAT IS AND IS NOT CLAIMED

Claimed. The no-go R1, with its $1.2\text{--}3.4 \times 10^4$ gap and the exhaustion of its candidate reliefs. The identity (5) and its corollary that AeST’s $c_4 = 0$ is a choice of variables. That $c_T = 1$ holds exactly in AeST and in the $\mathcal{F}(Z)$ theory for any free function. That the monotonicity obstruction of R1 *is* escapable, by the Z -form, whose background and $a_0(z)$ law survive exactly and whose CMB cost is met by the framework’s own derived suppression — conditionally on one uncomputed response.

Not claimed. That the Z -form *works*: its CMB pass is conditional on $S \lesssim 0.5\text{--}20$ and on two of eight forks that presently clash with the RAR ceiling, and neither δC_ℓ nor E itself has been computed. That $\kappa = \frac{1}{2}$ is derived; it is fitted, and must be quoted with its H_0 , the distance-free determination carrying an unpriced $\sim 7\%$ systematic. That dark matter is absent: Ω_{dm} is full here and the claim is “no dark-matter *particle*”. Whether

the dark sector’s dust remains bound inside galaxies is unresolved and untouched by any of this — it is a Q -sector problem, and all three constructions modify the $\mathcal{Y}$ -sector.

Unaffected by any of the above, being independent of which completion carries them: the normalisation (2), the radial acceleration relation at 0.108 dex on 175 SPARC galaxies, weak lensing from 40 kpc to 2.2 Mpc with no dark component, the baryonic Tully–Fisher relation, and a hash-frozen wide-binary prediction decided by Gaia DR4.

The one owed computation. A Boltzmann code carrying the unit-timelike A_μ , its constraint, and $\mathcal{F}(Z, Q)$, evolving the theory’s linear equations with K_B replaced by $K_B - \mathcal{F}_Z$ and reading off $S = |d \ln C_\ell / d\epsilon|$ directly. Every conditional statement in Sec. III collapses to a verdict once that number exists, and nothing else in this paper is waiting on anything. It is a build, not an argument.

A note on the process. Version 1 of this paper reported the Z -form as *killed* by a degeneracy $\mathcal{F}_Z(0) = K_B$. That was an error of ours: the source establishes that the analytic piece is degenerate *with* the K_B term in the sense of being indistinguishable from it — a relabelling — not that the equation degenerates. The correct value on the determined family is $\mathcal{F}_Z(0) = 0$. The error ran against our own interest and is recorded, dated, in RETRACTIONS.md. Superfluid constructions [13], in which the MOND force exists only inside a condensate phase, remain a separate candidate worth pricing.

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Every number in this paper is produced by a committed self-checking script at <https://github.com/carlzimmerman/zimmerman-formula>; each prints numbered checks and exits non-zero on failure. Withdrawn claims are recorded, dated, in RETRACTIONS.md — including one in this paper’s own drafting, where a gravitational-wave exclusion was written from an unverified number and corrected against the committed script before filing. The contribution here is the three requirements, the identity (5), and the constructions that establish them — not the underlying theory, which is Skordis and Złóśnik’s.

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